

THE LATHE & TURNING

The mother of machine tools — a rotating workpiece under a single-point cutting tool makes every shaft, pin, wheel, thread and engine part on Earth

	0.005 mm	finest turning/tolerance (precision/CNC)
	6,000+ rpm	spindle speeds on modern/high-speed turning centres
	Any metal	from soft aluminium to hardened 65 HRC steel
	12 axes	CNC turn-mill centres with live tooling + C-axis

THE BASICS

WHAT TURNING IS

Turning is the machining of a **rotating cylindrical workpiece** with a **single-point cutting tool** fed against it. Because the work spins about its own axis while the tool moves in straight lines, every feature produced is **rotationally symmetrical** — a perfect circle when measured around the axis, whatever the tool does.

- The **workpiece rotates** (driven at high speed by the spindle); the **tool stays still** or feeds linearly into it
- Cuts material as surface speed + feed slice a **chip** of set depth off the rotating diameter
- Produces **cylinders** (straight turning), **cones** (taper), **threads**, **bores**, **spheres** (form/CNC) and **faces**
- Parts: **shafts**, **pins**, **wheels**, **pulleys**, **bolts & threads**, engine **crankshafts**, **axles**, bearing journals, hubs, spindles
- The **lathe is the oldest machine tool** — a simple bow-driven wood lathe (~1300 BCE Egypt) predates every other machine by millennia
- Called the **mother of machine tools**: it can machine the parts to build every other machine (mill, drill, grinder) — even a copy of itself

Turning vs milling: on a lathe the **work turns** and the **tool moves** (circles about each axis); on a mill the **tool turns** and the **work moves** (flats, slots, 3D forms). Turning is fastest for anything round; milling for everything else.

WORKHOLDING

WORKHOLDING

How the work is gripped — every method

Choosing how to hold the work is the first decision of every job. Round stock suits a 3-jaw chuck; irregular castings need a 4-jaw or faceplate; long shafts run between centres.

- 3-jaw scroll chuck** — three jaws move together via a scroll plate → self-centring, fastest for round & hex stock; runout ~0.05–0.1 mm
- 4-jaw independent chuck** — each jaw dialled in separately → grips **square & irregular** work; can be set **true to 0.01 mm**
- Collets** (incl. ER chuck) — spring fingers grip the bar uniformly; **±0.005 mm** accuracy, no jaw marks, ideal for thin & bar stock
- Faceplate** — slotted plate; work bolted/strapped on → holds irregular castings & large flat jobs that can't be chucked
- Between centres** — dead/live centre in tailstock + **carrier (dog)** in the headstock drives it → most accurate for long shafts
- Mandrel** — a precise shaft pressed into the work's finished bore, so the outside turns **true to that bore**
- Steady rest & follower rest** — support long, slender work near the tool; stop deflection & chatter

Chuck types at a glance: scroll (self-centring, fast) · independent (accurate, awkward) · collet (precision) · combination (adjustable jaws) · magnetic/driving (special faceplate work).

MACHINE ANATOMY

ANATOMY — STRUCTURE (1/2)

The lathe is a rigid **cast-iron frame** whose parts align every cutting path to one axis. Each named part below has one job — learn them together and the whole machine makes sense.

- BED & WAYS**
- Bed** — heavy cast-iron base; keeps headstock, carriage & tailstock perfectly aligned; absorbs vibration & cutting forces
 - Ways** — hardened, ground guide rails (lit. 'V' or flat) the saddle slides along; must stay flat & parallel
- HEADSTOCK**
- Headstock** — left-hand housing for spindle, bearings, gears/belts and the speed (and feed) controls
 - Spindle** — hollow rotating shaft that drives the chuck or collet; the hollow bore lets long bar stock pass right through from the back
 - Chuck** — gripping jaws that hold the work; mounted on the spindle nose (threaded or cam-lock)

- TAILSTOCK**
- Tailstock** — right-hand slide, clamps anywhere along the bed; supports long work with a **centre**
 - Its quill** (Morse-taper socket) holds drills & reamers, or advances a **dead centre** into the work
 - Live centre** — rotating point for max accuracy at speed; **dead centre** — fixed, hardened point for light duty

OPERATIONS

TYPES OF TURNING OPERATIONS

Every cut the lathe can make

- All lathe work is a set of these operations — an engine shaft uses almost every one.
- Facing** — tool feeds across the end face to square it; the first operation after chucking
 - Straight (cylindrical) turning** — reduce diameter along the length; **roughing** (deep, fast) then **finishing** (light, fine feed, exact size)
 - Taper turning** — produce a cone by **swivelling the compound rest**, offsetting the tailstock, using a taper attachment, or CNC — e.g. Morse tapers ~1.5–3°
 - Parting-off & grooving** — a thin narrow tool cuts a groove (O-ring seat) or parts the finished piece off the bar
 - Boring** — single-point boring bar enlarges, straightens & trues a hole; corrects a hole drilled off-centre
 - Drilling & reaming** — tailstock quill feeds the drill in (centre-drill first so it can't wander); reamer finishes to size
 - Knurling** — hardened wheels press a diamond/straight grip pattern into the surface — handles & knobs; slow speed, light pressure, oil
 - Thread cutting** — single-point tool + leadscrew sync cuts 60° V-threads; **taps** cut internal, **dies** external threads
 - Chamfering** — bevel a sharp edge (debur & prep for assembly)
 - Profiling / form-tool turning** — a shaped tool or CNC arc cuts a curve, sphere or custom contour in one pass

Also possible: eccentric turning (work between two offset centres → crank pins & cams), hard turning (CBN on 45–65 HRC steel), and filing/polishing for finish.

CARRIAGE & DRIVE

ANATOMY — CARRIAGE & DRIVE (2/2)

- CARRIAGE**
- Carriage** — the assembly that rides the ways; carries the tool and feeds it longitudinally along the bed
 - Saddle** — the flat base plate riding directly on the ways (the bottom of the carriage)
 - Cross-slide** — moves the tool at exactly 90° to the axis → gives the **depth of cut** and does **facing**
 - Compound rest** — swivels to any angle on top of the cross-slide; set at an angle it produces **short tapers**; carries the toolpost
 - Toolpost / tool post** — clamps the tool bit (or quick-change holder) rigidly at **spindle centre height**
 - Apron** — the front face of the carriage; feed gears, clutch and the carriage handwheel, plus levers for forward/neutral/reverse feeds

- THE DRIVE — MAKING FEED & THREADS SYNCHRONISED**
- Leadscrew** — precision threaded shaft running along the bed; geared **1:1 with the spindle**, so carriage travel is locked to rotation — that shot of synchronisation is what cuts a perfect screw thread
 - Feed rod** — auxiliary shaft for normal power **turning/facing** feeds at selected rates, **without** thread-sync
 - Change gears** — interchangeable gears that set the spindle/leadscrew ratio, so **any thread pitch** can be cut
 - Power feeds**: **longitudinal** (along bed, for turning) and **cross** (across, for facing) — engaged from the apron

Follow the power: motor → gear/belt box → **spindle** (drives work) and, via change gears → **leadscrew/feed rod** (drives carriage) → **apron** feeds the tool. One motor, two perfectly-timed motions.

TOOLING

TOOLING

- TOOL MATERIALS**
- High-Speed Steel (HSS)** — tough, cheap, grindable; holds an edge to ~600°C; the manual/lathe default; cobalt-HSS for tough alloys
 - Carbide** — brazed or **indexable inserts**; 3–4x the speed of HSS; needs a rigid machine; the CNC default
 - Coated carbide** (TiN/TiAlN) — reduced friction, longer insert life
 - Ceramic** — cast iron & hardened steel at very high speeds
 - CBN** (cubic boron nitride) — hard turning, 45–65 HRC; **PCD** (polycrystalline diamond) — aluminium & composites

- TOOL GEOMETRY — THE ANGLES THAT CUT**
- Rake angle** — **positive** = sharp, light cuts (HSS); **negative** = strong edge, higher speeds (carbide)
 - Clearance / relief** ~5–10° — stops the flank rubbing and burning the work
 - Nose radius** — bigger = stronger edge + smoother finish but higher cutting forces
 - Cutting edge** — must sit exactly at **spindle centre height**, or it rubs/blows up or digs in

- HOLDERS & BARS**
- Tool holders** — quick-change, right/left-hand and straight shanks clamp the bit or insert
 - Boring bars** — long, cantilevered bars that reach into holes; overhang = flex = chatter, so keep them short and stiff
 - Indexable insert code **CNMG 12 04 08**: C=80° rhombus shape · N=0° relief · M=tolerance · G=chipbreaker · 12=size · 04=thickness · 08=0.8 mm nose radius

SPEEDS & FEEDS

SPEEDS, FEEDS & DEPTH OF CUT

Cutting speed is set by the material & tool → not by the dial

Cutting speed (CS, V) is the surface speed of the work in m/min, chosen for the material–tool pair. Spindle **RPM** = **(1000 × V) ÷ (π × D)** where D = diameter in mm. Feed rate (mm/min) = feed (mm/rev) × RPM. Depth of cut = the radial layer removed per pass.

Example: Ø40 mm mild steel, HSS, CS = 35 m/min → RPM = 35,000 ÷ 125.7 = 278 rpm; with carbide (CS 140) → = 1,114 rpm. Smaller Ø spins faster; bigger Ø slower, to keep surface speed constant.

Workpiece material	HSS cutting speed (m/min)	Carbide cutting speed (m/min)	Feed rough (mm/rev)	Feed finish (mm/rev)
Mild / low-carbon steel	30–45	90–180	0.2–0.5	0.05–0.15
Aluminium alloys	60–120	200–400	0.15–0.4	0.05–0.15
Brass & bronze	45–75	150–250	0.15–0.4	0.05–0.15
Cast iron	20–30	60–120	0.2–0.5	0.05–0.15
Stainless steel	15–25	60–120	0.15–0.35	0.05–0.12
Plastics	60–120	200–300	0.1–0.3	0.05–0.1

Rules of thumb: depth of cut — roughing 1–4 mm (HSS) / 2–8 mm (carbide); finishing 0.1–0.5 mm. Chatter = too fast/deep, dull tool or too much overhang → slow down. Threading: several light passes, finish with a zero-cut spring pass. Coolant: soluble oil (steel) · paraffin (aluminium) · dry (cast iron). **Material removal rate (MRR, cm³/min)** = **cutting speed × feed × depth of cut** — the product you're maximising in production.

AUTOMATION

CNC LATHE / TURNING CENTRE

A **CNC turning centre** replaces the machinist's hand on the wheel and the screw-cutting gear train with a **computer driving servo axes** — holding ±0.001 mm and running unattended overnight. The program (G-codes) commands spindle speed, axis feeds and tool changes.

- Horizontal vs vertical (VTL)**: horizontal spindles suit long shafts & bar work; vertical lathes sit heavy, large-diameter discs & flywheels flat on a faceplate so gravity holds them
- Turret tooling** — a multi-tool turret (8–12 stations) indexes the right tool in seconds → one cycle does rough, finish, groove, thread
- Live tooling** — rotating (driven) tools let the centre do **milling**, drilling & cross-hole work without a second machine; with a **C-axis** spindle it becomes a full **turn-mill**
- Bar feeder** — automatic magazine feeds bar stock through the hollow spindle, so parts are made and parted off continuously, lights-out
- Tailstock** on CNC supports long shafts; some centres add a **sub-spindle** that grabs the part to machine its back face in one setup
- CNC-to-12-axis turn-mill centres combine rotating & stationary tooling, sub-spindles and Y/B axes for hugely complex parts in a single clamping

Vs the manual engine lathe: CNC = fast, precise, repeatable, expensive, best for **production**; manual = cheap, flexible, skill-driven, best for **one-offs, repair & learning** — still the first machine every machinist learns on.

TROUBLESHOOTING

DEFECTS & CHATTER

The problems — recognise & fix

Every bad lathe result has a cause; every cause has a fix.

- TOOL CHATTER / VIBRATION**
- Wavy chatter marks on the surface + a loud 'squeal'
 - Caused by: too fast/deep a cut, **dull tool**, **too much overhang** (work or boring bar), insufficient rigidity, work not firmly gripped
 - Fix: slow down / lighten the cut, sharpen or change the tool, support with a steady rest, shorten the overhang, take a spring cut
- POOR FINISH**
- Rough, torn or burnt surface — wrong feed/speed, dull edge, tool 'rubbing' on clearance, no coolant
 - Fix: reduce feed, raise/lower speed to the sweet spot, sharpen & correct the angles, use coolant, check tool at centre height
- TAPER ERRORS**
- Work is slightly conical when it should be parallel — headstock tailstock **misaligned**, work deflects from a heavy cut, or tailstock centre offset
 - Fix: align tailstock (test between centres), lighter cuts, use a centre or steady rest
- BUILT-UP EDGE (BUE)**
- Work material welds to the cutting edge, giving a rough, smeared surface & poor accuracy — common on soft aluminium & low-carbon steel at low speed
 - Fix: raise cutting speed, increase rake, use coolant/lubricant, sharper ground edge
- OVERHEATING / BURNT WORK**
- Discoloured, softened or welded chips — too fast/thin a cut, no coolant, wrong tool material
 - Fix: more coolant flow, reduce speed, use a tougher edge, deeper cuts to get under the hot skin

TRADE-OFFS

ADVANTAGES vs LIMITATIONS

Strengths, limits & the machine comparison

- WHY CHOOSE A LATHE**
- ✓ Perfect roundness & concentricity — symmetric about the axis
 - ✓ Machines shafts, threads, cones & bores nothing else matches
 - ✓ Excellent, repeatable finish & ±0.01 mm (manual) to ±0.001 mm (CNC) accuracy
 - ✓ One setup can rough, finish, thread & part a complete shaft
 - ✓ Cuts any machinable material — HSS to diamond tooling
 - ✓ The tool is cheap & simple (a single cutting edge)
- LIMITATIONS**
- ✗ Only produces rotationally symmetric parts
 - ✗ Cannot cut internal square corners, flats or 3D forms on its own (needs live tooling/mill)
 - ✗ Long slender work deflects without extra support
 - ✗ Manual lathes are slower & operator-skilled for batch work
 - ✗ Can't easily make a part already mounted off-axis

Machine	Work moves	Tool moves	Best at	Limit
Lathe	rotates	feeds linearly	shafts, threads, bores	symmetric parts only
Mill	translates (X,Y,Z)	rotates	flats, slots, pockets, 3D	no true turned circles
CNC router	large flat sheets	rotates (3-axis+)	2D/3D shapes in sheet/soft mat.	not for round work

STAY SAFE

LATHES SAFETY — RULES THAT SAVE FINGERS

The rules are straightforward — these habits prevent most serious injuries

Rule	Why it matters
No gloves around the rotating chuck	A glove is grabbed, wraps and pulls your hand & arm in — a classic, lethal injury
Tie back long hair, no loose clothing or jewellery	Loose material wraps around the spindle and drags you onto the work
Remove the chuck key before starting	A thrown chuck key is a lethal missile — a leading cause of lathe injury
Safety glasses always	Hot swarf, coolant and broken carbide fly at high speed
Stop the spindle to measure, adjust or change tools	Measuring on a moving chuck drags fingers in; always brake to a full stop
Work must be firmly gripped	A loose part spins off or is flung across the shop; re-tighten after the first roughing cut
Clear swarf with a brush/brush, never bare hands	Fresh swarf is sharp and hot — it will cut and burn
Never slow the chuck with your hand — use the brake	Grabbing the spinning chuck burns and breaks fingers
Keep guards closed & know the E-stop	Guards deflect swarf & coolant; the E-stop is your last-resort stop
Do not run too fast for the job	Excess speed causes chatter, tool breakage and thrown work

VOCABULARY

GLOSSARY — KEY LATHE TERMS

10 terms that tie the whole machine together

- Bed** — heavy cast-iron base aligning all parts & absorbing vibration
- Spindle** — hollow rotating shaft driving the chuck/collet
- Carriage** — assembly riding the ways that feeds the tool
- Cross-slide** — moves the tool 90° to the axis (depth & facing)
- Chuck** — gripping jaws holding the work on the spindle nose
- Centres** — dead (fixed) & live (rotating) 60° points for between-centres work
- Feed rod** — auxiliary shaft for normal turning/facing power feeds
- Knurling** — hardened wheels pressing a diamond/straight grip pattern
- Toolpost** — clamps the tool bit at spindle centre height
- Headstock** — houses the spindle, bearings, gears & speed controls
- Tailstock** — right-hand support for long work & drill holding
- Saddle** — flat base of the carriage riding directly on the ways
- Compound rest** — swivelling slide on the cross-slide for tapers
- Collet** — spring fingers gripping bar stock precisely (±0.005 mm)
- Leadscrew** — precision threaded shaft, geared 1:1, for screw cutting
- Taper** — a gradual cone — Morse tapers ~1.5–3°, set by compound, offset or CNC
- Apron** — front of carriage: feed gears, clutch & handwheel
- Mandrel** — precision shaft in the bore so the outside turns true to it

QUICK FACTS

THE LATHE IN ONE LINE

- "Mother of machine tools"** — the only machine tool that can build a copy of itself.
- ~1300 BCE — first bow-driven wood lathes in Egypt
 - 1797 — Maudslay's screw-cutting lathe: standardised threads & interchangeable parts, the seed of mass production
 - Every engine, gearbox, axle, bolt, bearing & screw on Earth is shaped by a lathe or by machines the lathe built
 - Manual accuracy ±0.01 mm · CNC turning centres ±0.001 mm
 - Watch screws from Swiss lathes are thinner than a human hair; the largest lathes turn multi-metre ship shafts
- Evolution: bow lathe (~1300 BCE) → Roman two-man → pole lathe → Da Vinci's flywheel treadle (~1480) → Besson screw cutting (1568) → Maudslay all-metal (1797) → turret lathes (1840s) → geared engine lathe (1890s) → punched-tape NC (1950s) → CNC turning centres (1970s–).

EXAM TIP

DESIGN & TECHNOLOGY — ANSWERING LATHE QUESTIONS

Describe how a part is turned on a lathe → 1. chuck the work, 2. select speed & feed for the material, 3. face the end, 4. turn to diameter with roughing then finishing cuts, 5. measure with micrometer, 6. part off.

FACING vs TURNING: tool feeds across the end vs along the length. ROUGHING vs FINISHING: deep fast cuts leave stock, then light fine cuts hit size. CENTRES: use them for long, accurate shafts. CHATTER: too fast/deep, dull tool or overhang — slow down.

Quick revision: the leadscrew is geared 1:1 with the spindle to cut threads · a swivelled compound rest cuts a taper · a collet gives ±0.005 mm · carbide runs 3–4x faster than HSS · never wear gloves near a rotating chuck.